

ABDULLAH MD JAHID HASSAN

Backend Software Engineer — Python · FastAPI · REST APIs · PostgreSQL · Docker

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Languages: Bangla (native) · English (professional) · Hindi (conversational)



PROFESSIONAL SUMMARY

Backend engineer who builds and maintains **backend services and REST APIs using FastAPI and Python** — four production FastAPI services and four on Django REST Framework, delivered across eleven months. Currently **Senior Python Executive at Softvence Agency**, promoted from Junior Python Developer within eleven months on end-to-end delivery rather than tenure. Day-to-day work is the shape this role describes: **Pydantic** schemas over **async** Python, **PostgreSQL** data models and efficient queries, **REST API design with JWT/OAuth2 authentication**, **background jobs** on Celery and Celery Beat, and **API integrations** with Stripe, OpenAI, Twilio, Firebase and n8n. Owns each service through to production — **Docker**, **Git**, **pytest**, GitHub Actions CI/CD and deployment on AWS EC2. A fast, self-directed learner by record rather than by claim: moved from PHP/MySQL to production Django inside a year, then picked up FastAPI and shipped four services on it.

TECHNICAL SKILLS

Python & FastAPI: Python 3.11–3.13 (primary) · **FastAPI 0.119** · **Pydantic v2** + pydantic-settings · **async**/await, asyncio, async application-factory and lifespan patterns · Uvicorn · async ORM: **Tortoise ORM 0.25** + **Aerich migrations** · async drivers (asyncpg, aiomysql) · python-socketio · APScheduler

Django & Django REST Framework: Django 5.2 · Django REST Framework 3.16 · SimpleJWT · django-allauth / dj-rest-auth · django-filter · django-storages · Django Channels (WebSockets) · Gunicorn

PostgreSQL & data modelling: **PostgreSQL** (primary; psycopg3, asyncpg) · MySQL · SQLite · schema design, normalisation and ER modelling · **efficient queries** — composite indexing, select_related/prefetch_related, N+1 elimination, bulk_update over per-row saves · database constraints and custom managers · migrations and migration-drift gating

REST APIs & authentication: **REST API design** and versioning · **JWT** (SimpleJWT, python-jose, PyJWT) with refresh-token rotation and blacklisting · **OAuth2** / Google OAuth 2.0 · TOTP two-factor via pyotp · RBAC and permission groups · rate limiting and throttling · unified response envelope · Swagger / OpenAPI via drf-yasg

Background jobs & caching: **Celery 5** + **Celery Beat** with isolated queue routing · **Redis 7** as cache, task broker, OTP store and WebSocket channel layer · scheduled and cron-driven jobs · idempotent webhook processing · retry, watchdog and dunning state machines

Docker, Git & testing: **Docker** (multi-stage builds) · Docker Compose · **Git** / GitHub (branching, pull-request workflows, **code review**) · **pytest** + pytest-asyncio · Django TestCase · Bruno and Postman collections · Locust · Sentry · GitHub Actions CI/CD · AWS EC2, IAM, S3 · Nginx · Caddy · Linux

API integrations: Stripe (subscriptions, invoices, webhooks) · Tap Payments · OpenAI · ElevenLabs · Twilio · **Firebase Cloud Messaging** (push notifications) · Cal.com · Google Calendar · n8n · AWS SES / SMTP

PROFESSIONAL EXPERIENCE

Senior Python Executive — Softvence Agency

Aug 2025 – Present

Mohakhali, Dhaka · Full-time, on-site · Promoted from Junior Python Developer, Jul 2026

- **Developed and maintained eight production backend services and their REST APIs** in eleven months — **four on FastAPI** and four on Django REST Framework — each owned end to end from data model through API, containerisation and live deployment.
- **Promoted from Junior Python Developer to Senior Python Executive within eleven months** on the strength of end-to-end ownership rather than tenure.
- **Own well-defined modules end to end on shared codebases** — on a three-engineer team I held the authentication, subscription, OTP, notification and logging domains while colleagues held theirs, integrating through reviewed interfaces.
- **Design PostgreSQL data models and write the queries that read them efficiently** — normalised schemas with database-level uniqueness constraints and composite indexes on every query path, N+1 elimination via select_related/prefetch_related, and bulk writes in place of per-row saves.
- **Build background jobs as first-class infrastructure** — Celery and Celery Beat queues with explicit, isolated routing keys so slow jobs cannot block business tasks, plus scheduled watchdogs, renewal tasks and retry state machines backed by Redis.
- **Contribute to API integrations end to end** — Stripe, Tap Payments, OpenAI, ElevenLabs, Twilio, Cal.com, Firebase Cloud Messaging and n8n, each behind an idempotent webhook or authenticated callback contract so replays and retries are safe.
- **Participate in code reviews, testing and debugging** — pull-request review on shared repositories, pytest and Django TestCase suites, Bruno and Postman collections for API regression, and a CI pipeline that gates on migration drift and runs the test suite inside the built image rather than beside it.
- **Collaborate with frontend, QA and product** — separate frontend teams on most projects, with requirement analysis taken directly from client calls and briefs and returned as versioned specifications, API contract documentation and per-app developer guides.

Founder, Lead Engineer & Designer — Omnyvora

Jun 2026 – Present

Own organisation · github.com/orgs/omnyvora · Sole contributor

- **Took a multi-tool SaaS platform from a blank repository to a running stack alone** — a Django REST modular monolith across four bounded domains with strict models → selectors → services → serializers → views layering, so views never touch the ORM directly and each module stays independently testable.

- **Modelled normalised personnel records** — profiles, skills, experience, education and certifications — with validated fields, ownership checks that reject a whole payload if any record is not the caller's, and reordering served by a single `bulk_update` rather than per-row saves.

Web Development Intern — Cyber Hundred

Feb 2024 – Jul 2024

Khilkhet, Dhaka · On-site · Certificate issued Jun 2024

- Built frontend and backend features in HTML5, CSS, Bootstrap, PHP and MySQL across multiple client web projects; managed databases through phpMyAdmin; used Git and GitHub for collaborative development.

SELECTED PROJECTS

OCReels — Async FastAPI Media Platform · backend developer

Jun 2026 – Jul 2026

ocreels.com · FastAPI 0.119 · Pydantic v2 · Tortoise ORM + Aerich · PostgreSQL (asyncpg) · Redis · Celery · pytest-asyncio · Docker · AWS EC2

- **Built on a fully asynchronous, auto-discovering FastAPI architecture** — an application factory with an async lifespan hook that initialises database, Redis, scheduler and permission sync, then dynamically registers routes and auto-imports each module's signals, so adding a domain needs no change to `main.py`.
- **Implemented the creator programme as an application** → review → approval state machine over an earnings ledger keyed by source and content, with reviewer attribution, rejection reasons and hashed TOTP backup codes — the same review-and-approve shape an HR workflow runs on.

ArchiCoPro — Members' Portal · lead backend developer

Jun 2026 – Jul 2026

archicopro.cloud · FastAPI · Tortoise ORM · PostgreSQL · Redis · Celery · Docker Compose · Caddy · GitHub Actions

- **Designed feature-level access control where permissions are granted per role and overridden per person** — a feature enum with role permission tables plus per-user overrides, enforced per resource rather than per endpoint, so a document folder, article or training is visible only to the people entitled to it. This is the permission model an HRIS needs to keep employee records private.
- **Built the member-record and lifecycle domains** — a typed user status lifecycle, session and activity auditing with a typed action enum, registration flows for events and trainings, and a moderated-forum state machine with a full moderation audit trail.

CHYR — Multi-Tenant B2B SaaS Platform · backend developer

Apr 2026 – May 2026

Python · Django 5.2 · DRF · PostgreSQL · Redis · Celery + Beat · Stripe · Docker · AWS EC2 (two regions)

- **Modelled the organisation, membership and onboarding domain** — one organisation per owner with role-scoped member profiles, and an invitation flow issuing expiring unique tokens whose acceptance validates the password, creates the user and provisions the profile inside a single atomic transaction, so a half-created member cannot exist.
- **Built the subscription and metered-entitlement engine** — plans composed of features with usage limits, a Celery renewal task applying scheduled plan changes at the next billing cycle, and idempotent Stripe webhook processing keyed on a unique event ID.

ProspectLead — Credit-Based SaaS · lead backend developer

Jan 2026 – Jul 2026

prospectlead.ai · Django · Django REST Framework · PostgreSQL · Redis · Celery + Beat · Stripe · n8n · Bruno

- **Owned the billing engine end to end** — an append-only ledger carrying a running-balance snapshot per row, and reserve-on-submit / finalise-on-callback semantics so concurrent jobs cannot overspend a balance.
- **Made an external, long-running pipeline safe to depend on** — a bearer-token-authenticated inbound callback contract plus a **Celery Beat watchdog** that auto-fails and refunds any request still unanswered after six hours, so a silent third-party failure never strands a user's credits.

Also delivered — **Tundra** (thundra.de · three-service Django + FastAPI e-commerce estate) · **Outdoorda** (FastAPI two-sided marketplace) · **LookUp** (Django REST gamification backend) · **django_templet**, my own reusable Django REST starter with Celery, Docker and CI/CD built in.

EDUCATION & CERTIFICATIONS

B.Sc. in Computer Science & Engineering

May 2020 – Jun 2024

IUBAT — International University of Business Agriculture and Technology, Dhaka · CGPA 3.59 / 4.00 · Degree certificate issued 30 May 2024

- **Thesis:** *Implementation of Parallel Data Transmission in Li-Fi Technology Using RGB Light*. **Coursework:** Data Structures & Algorithms · DBMS · Software Engineering · Computer Networks.

Python-Django — EDGE, Enhancing Digital Government and Economy (Bangladesh Government ICT programme) · Jul 2025 · **Internship Certificate** — Cyber Hundred · Jun 2024 · References available on request.

HOW I WORK

Every service I ship is containerised with a multi-stage **Docker** build, gated by a **GitHub Actions** pipeline that fails on migration drift and runs `pytest` inside the built image, then deployed to AWS EC2 or a VPS behind Nginx or Caddy. Each repository carries written developer guides and API documentation so the next engineer can pick it up without me, plus a hand-written `CONTEXT.md` encoding the architectural rules my daily **Claude Code** and **Cursor** sessions must follow — architecture, data modelling and code quality stay mine.